

EE 3TR4 (COMMUNICATION SYSTEMS): MIDTERM EXAMINATION
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Student Name: _____

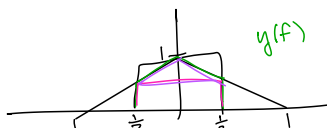
Student Number: _____

- 1) The impulse response of a filter is $h(t) = \text{sinc}(t)$ and its input is $x(t) = \text{sinc}^2(t)$. Find its output $y(t)$. 5 marks

$$y(f) = \begin{cases} 1-|f| & , |f| \leq 0.5 \\ 0 & , \text{else} \end{cases}$$

$$y(f) = \frac{1}{2} \text{rect}(f) + \frac{1}{2} \text{triang}(2f) \rightarrow y(t) = \frac{1}{2} \text{sinc}(t) + \frac{1}{2} \cdot \frac{1}{2} \text{sinc}^2\left(\frac{t}{2}\right)$$

$$= \frac{1}{2} \text{sinc}(t) + \frac{1}{4} \text{sinc}^2\left(\frac{t}{2}\right)$$



- 2) Consider an AM wave for the following message signal:

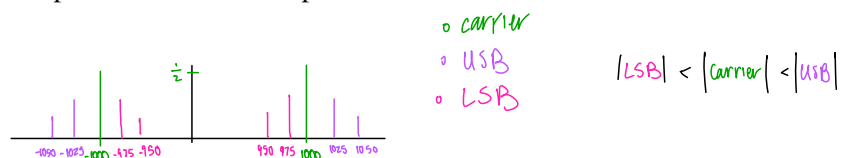
$$m(t) = 2 \sin(50\pi t) + \cos(100\pi t).$$

The AM wave is

$$s(t) = [1 + k_a m(t)] \cos(2000\pi t)$$

with $k_a = \frac{1}{4}$.

- a) Find the Fourier transform of the AM wave, $S(f)$, sketch the magnitude spectrum, $|S(f)|$, and identify the upper and lower sidebands. 3 marks
- b) Find the ratio of the upper sideband power to the carrier power. 3 marks



- 3) Envelope detection:

- a) If the message signal is $m(t) = \cos(100\pi t)$ and the carrier frequency is $f_c = 10^5 \text{ Hz}$, identify the feasible range of the time constant of the output circuit, $R_L C$, and sketch the output of the envelope detector (assuming % modulation = 50%).

4 marks

- b) If an DSB-SC signal with the message signal $m(t) = \cos(100\pi t)$ is applied to an ideal envelope detector, sketch the output of the envelope detector. Can the message signal be retrieved without distortion?

4 marks

Fourier transform of special functions:

$$\begin{aligned}\text{rect}(t) &\xleftrightarrow{\mathcal{F}} \text{sinc}(f), \quad \text{triang}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}^2(f), \\ \cos(2\pi f_c t) &\xleftrightarrow{\mathcal{F}} \frac{1}{2}[\delta(f - f_c) + \delta(f + f_c)], \quad \sin(2\pi f_c t) \xleftrightarrow{\mathcal{F}} \frac{1}{2j}[\delta(f - f_c) - \delta(f + f_c)],\end{aligned}$$

where

$$\begin{aligned}\text{rect}(t) &= \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & \text{otherwise,} \end{cases} \\ \text{triang}(t) &= \begin{cases} 1 - |t|, & |t| < 1, \\ 0, & \text{otherwise,} \end{cases} \\ \text{sinc}(t) &= \frac{\sin(\pi t)}{\pi t}.\end{aligned}$$

Properties of Fourier transform:

If $g(t) \xleftrightarrow{\mathcal{F}} G(f)$ and $h(t) \xleftrightarrow{\mathcal{F}} H(f)$, then

$$\begin{aligned}g(at) &\xleftrightarrow{\mathcal{F}} \frac{1}{|a|} G\left(\frac{f}{a}\right), \\ G(t) &\xleftrightarrow{\mathcal{F}} g(-f), \\ g(t - t_0) &\xleftrightarrow{\mathcal{F}} G(f) e^{-j2\pi f t_0}, \\ g(t) e^{j2\pi f_c t} &\xleftrightarrow{\mathcal{F}} G(f - f_c), \\ g(t) \cos(2\pi f_c t) &\xleftrightarrow{\mathcal{F}} \frac{1}{2} [G(f - f_c) + G(f + f_c)], \\ g(t) * h(t) &\xleftrightarrow{\mathcal{F}} G(f) H(f), \\ g(t) h(t) &\xleftrightarrow{\mathcal{F}} G(f) * H(f).\end{aligned}$$

Amplitude modulation:

Standard AM: $s(t) = A_c[1 + k_a m(t)] \cos(2\pi f_c t)$.

DSB-SC: $s(t) = A_c m(t) \cos(2\pi f_c t)$.

Trigonometric identities:

$$\begin{aligned}2 \cos(A) \cos(B) &= \cos(A - B) + \cos(A + B), \\ 2 \sin(A) \sin(B) &= \cos(A - B) - \cos(A + B), \\ 2 \sin(A) \cos(B) &= \sin(A - B) + \sin(A + B).\end{aligned}$$

Mean power of a sinusoid: The mean power of $A \sin(2\pi f t + \theta)$ is $\frac{A^2}{2}$.

Solution to the midterm

1) Since $H(f) = \text{rect}(f)$ and $X(f) = \text{triang}(f)$, it follows that

$$Y(f) = X(f)H(f) = \text{rect}(f)\text{triang}(f) = \frac{1}{2}\text{rect}(f) + \frac{1}{2}\text{triang}(2f).$$

Therefore, we have

$$y(t) = \frac{1}{2}\text{sinc}(t) + \frac{1}{4}\text{sinc}^2\left(\frac{t}{2}\right).$$

2) a) Note that

$$M(f) = j[\delta(f + 25) - \delta(f - 25)] + \frac{1}{2}[\delta(f - 50) + \delta(f + 50)].$$

Consequently,

$$\begin{aligned} S(f) &= \frac{1}{2}[\delta(f - 1000) + \delta(f + 1000)] + \frac{k_a}{2}[M(f - 1000) + M(f + 1000)] \\ &= \frac{1}{2}[\delta(f - 1000) + \delta(f + 1000)] \quad \text{carrier} \\ &\quad + \frac{j}{8}[\delta(f + 1025) - \delta(f - 1025)] + \frac{1}{16}[\delta(f - 1050) + \delta(f + 1050)] \quad \text{upper sideband} \\ &\quad + \frac{j}{8}[\delta(f - 975) - \delta(f + 975)] + \frac{1}{16}[\delta(f - 950) + \delta(f + 950)] \quad \text{lower sideband.} \end{aligned}$$

The magnitude spectrum, $|S(f)|$, can be sketched as follows: two pulses of height $\frac{1}{2}$ at $\pm 1000\text{Hz}$, two pulses of height $\frac{1}{8}$ at $\pm 1025\text{Hz}$, two pulses of height $\frac{1}{16}$ at $\pm 1050\text{Hz}$, two pulses of height $\frac{1}{8}$ at $\pm 975\text{Hz}$, and two pulses of height $\frac{1}{16}$ at $\pm 950\text{Hz}$.

b) The mean power of the upper sideband signal $\frac{1}{4}\sin(2050\pi t) + \frac{1}{8}\cos(2100\pi t)$ is $\frac{1}{2}\left[\left(\frac{1}{4}\right)^2 + \left(\frac{1}{8}\right)^2\right] = \frac{5}{128}$ while the mean power of the carrier $\cos(2000\pi t)$ is $\frac{1}{2}$. Therefore, the ratio of the upper sideband power to the carrier power is $\frac{5}{64}$.

- 3) a) The bandwidth of the message signal is $W = 50\text{Hz}$. We require $\frac{1}{f_c} \ll R_L C \ll \frac{1}{W}$, i.e., $10^{-5}\text{s} \ll R_L C \ll 0.02\text{s}$. The output of the envelope detector is approximately $1 + \frac{1}{2}\cos(100\pi t)$.
- b) The output of the envelope detector is approximately $|\cos(100\pi t)|$. The message signal cannot be retrieved without distortion.